

Yamini E

Sr. Python Developer | AI/ML Engineer

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PROFESSIONAL SUMMARY

Dynamic and results-driven Python Developer with 6 years of experience delivering high-quality enterprise solutions across healthcare, industrial, and e-commerce sectors. Specialized in designing and developing robust Python-based microservices, AI/ML pipelines, and scalable cloud-native architectures using AWS and Azure. Hands-on expertise in building intelligent data-driven products including predictive analytics, NLP models, and real-time processing systems. Adept at modern frontend frameworks (React, Angular), CI/CD automation, and infrastructure as code practices. Strong background in optimizing data workflows, integrating advanced machine learning algorithms, and delivering highly secure, performant, and user-centric applications. Proven ability to lead feature development from concept to deployment, mentor teams, and align technology solutions to business objectives.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, TypeScript

Frameworks & Libraries: Django, Flask, FastAPI, TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, XGBoost, Hugging Face Transformers

Frontend: React.js, Angular, Vue.js, HTML5, CSS3, Bootstrap, Chart.js

Cloud & DevOps: AWS (S3, EC2, Lambda, SageMaker, Bedrock), Azure (Functions, Blob, Data Lake), GCP (Vertex AI, Pub/Sub), Docker, Kubernetes, Terraform, Jenkins, GitHub Actions.

Databases: SQL Server, Oracle, MongoDB, SSIS, Pandas, Scikit-learn, Numpy, NLP, CNN, DBN

Visualization & Reporting: Tableau, Power BI, Seaborn, Matplotlib

Tools: Git, Jira, Confluence, Postman, Jupyter Notebook, VS Code, PyCharm

AI/ML: NLP, CNN, RNN, LSTM, RAG pipelines, MLOps, Feature Engineering, Model Deployment

Methodologies: Agile (Scrum), TDD, CI/CD, Data Wrangling, ETL, Microservices

COURSES & CERTIFICATES

- Learned about Attacks on a system, DoS, DDoS, session hijacking, web server and web application attacks, SQL injection, wireless, and SQL injection methodology and evasion techniques Hands on Python and R in Data Science UDEMY
- Learned to make accurate predictions and which machine learning model to choose for each type of problem
- Prediction and Comparative Analysis of Air Pollution in Major cities of India using Deep Learning Techniques

PROFESSIONAL EXPERIENCE

Client: Google (CA)

Aug 2025 - Present

Role: Python Developer with AI/ML

- Designed and implemented Python-based microservices supporting HIPAA-compliant healthcare workflows, including patient scheduling and care-plan optimization.
- Developed predictive readmission models (XGBoost, Logistic Regression) integrated with patient discharge systems, improving planning accuracy by 30%.
- Built NLP-driven patient feedback analytics using spaCy and Hugging Face, enhancing sentiment tracking for clinical decision support.
- Orchestrated ETL pipelines with PySpark and Pandas to ingest and transform EHR and claims data for real-time analytics dashboards.
- Created AI-powered recommendation engines for medication plans, dynamically personalized for each patient's medical history.
- Integrated AWS SageMaker pipelines for continuous model training and automated deployment, reducing manual intervention by 40%.
- Leveraged RAG pipelines to summarize lengthy clinical documents, cutting physician reading time by 50%.
- Deployed scalable microservices with FastAPI and Docker, managed on Kubernetes clusters for seamless horizontal scaling.
- Built real-time anomaly detection systems for lab result data streams, enabling proactive clinical alerts and faster intervention.

- ❑ Implemented patient portal chatbots using GPT models, automating routine inquiries and reducing support workload.
- ❑ Designed data encryption workflows with AWS KMS and column-level encryption, ensuring compliance with PHI standards.
- ❑ Constructed dynamic care-team dashboards using React.js and Plotly for visualizing patient risk scores and treatment KPIs.
- ❑ Integrated Kafka-based event streams to synchronize real-time updates across clinical systems and partner networks.
- ❑ Implemented automated feature engineering pipelines, standardizing model input data and reducing preprocessing errors.
- ❑ Designed AI-driven care gap closure modules, improving preventive care compliance metrics.
- ❑ Developed robust API testing frameworks using PyTest, achieving 90%+ test coverage across service endpoints.
- ❑ Automated data quality audits using Python scripts and Azure Data Factory, improving data integrity for downstream analytics.
- ❑ Configured RBAC and granular IAM policies for ML endpoints, securing sensitive prediction services.
- ❑ Led migration of legacy on-prem analytics jobs to cloud-native AWS Glue ETL, accelerating report generation cycles by 60%.
- ❑ Mentored junior engineers on Python best practices, model interpretability techniques, and cloud deployment strategies.
- ❑ Collaborated with product and compliance teams to integrate model explainability modules, enhancing audit readiness.
- ❑ Enabled real-time cohort analysis for population health teams, supporting targeted outreach and intervention strategies.
- ❑ Developed VectorDB-based semantic search for internal medical research documents, reducing time to insights for clinicians.

Client: Shrewin Williams (GA)

Jan 2024 - Jul 2025

Role: Sr. Python Developer with AI/ML

- ❑ Engineered Python-based microservices to process and optimize 3D CAD models for additive manufacturing workflows.
- ❑ Developed real-time telemetry data ingestion and analysis modules using FastAPI and PySpark, enabling proactive printer maintenance.
- ❑ Integrated predictive maintenance ML models (Random Forest, XGBoost) to reduce unplanned printer downtime by 35%.
- ❑ Built Flask APIs for managing job queues, printer health status, and maintenance logs, exposing secure endpoints to internal applications.
- ❑ Designed visualization dashboards in Angular and Plotly for monitoring print progress, error rates, and material usage.
- ❑ Integrated containerized ML inference services on AWS SageMaker and deployed via EKS for scalable, low-latency predictions.
- ❑ Automated print job scheduling using event-driven architecture with Kafka and Celery task queues.
- ❑ Implemented advanced anomaly detection on sensor data streams to flag out-of-spec prints and initiate auto-correction routines.
- ❑ Built secure OAuth2.0 and JWT-based authentication mechanisms for controlling internal access to 3D printing APIs.
- ❑ Developed data lake integration with AWS S3 to store and analyze historical print data, supporting design optimization.

- Engineered asynchronous feedback loops using WebSockets for real-time operator notifications and printer health alerts.
- Implemented GraphQL API gateways to aggregate printer metadata and provide flexible client query interfaces.
- Developed ML-based topology optimization tools for additive parts, reducing material costs and improving mechanical properties.
- Enhanced CI/CD pipelines with automated model retraining triggers, reducing manual oversight and improving system resilience.
- Built internal quality assurance tools using Python and TensorFlow to evaluate layer-by-layer print consistency.
- Led knowledge-sharing sessions on AI integration with industrial automation, improving team capabilities across global teams.
- Optimized data preprocessing steps using NumPy and Dask, achieving significant reductions in model training times.

Company: IBM/Kyndryl (INDIA)
Role: Python Developer with AI/ML

Oct 2019 – Jul 2023

- Developed robust Python microservices for e-commerce order processing, payment reconciliation, and inventory tracking.
- Engineered recommendation engines using collaborative filtering and product similarity models, boosting cross-sell revenue by 25%.
- Built React-based frontend modules for dynamic product detail pages, integrating personalized offers and real-time stock updates.
- Integrated Elasticsearch for fast, semantic product search and dynamic category filtering capabilities.
- Developed Flask APIs to manage promotions, flash sales, and time-bound discount logic, reducing manual campaign overhead.
- Implemented AI-driven customer churn prediction models using Scikit-learn, enabling targeted re-engagement marketing.
- Developed analytics dashboards in Vue.js and Chart.js to visualize sales trends, conversion rates, and campaign performance.
- Automated A/B testing workflows for promotional features, collecting metrics and feeding insights into optimization cycles.
- Configured Dockerized services and orchestrated deployments on Kubernetes, ensuring zero-downtime during major sale events.
- Built session analysis modules using TensorFlow to track user behavior patterns, supporting UI/UX optimization strategies.
- Integrated with third-party logistics APIs for live shipment status and automated customer notifications.
- Implemented real-time fraud detection pipelines using anomaly scoring on transaction data streams.
- Optimized database queries and caching strategies for high-traffic periods, reducing checkout timeouts and abandonment rates.
- Collaborated with design and product teams to implement accessibility features, achieving WCAG 2.1 compliance.
- Documented API specifications with Swagger and led code reviews to enforce best practices and maintain code quality.

EDUCATION

Master of Science in Computer Science from University Of Central Missouri – USA(2024)

Bachelor's in Computer Science from Audisankara Institute Of Technology – INDIA(2019)